

Topics

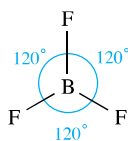
Symmetry operators
Symmetry elements
Point groups
Character tables
Vibrational modes in molecules
Chiral molecules

3

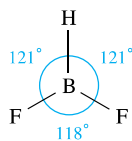
Introduction to molecular symmetry

3.1 Introduction

In chemistry, symmetry is important both at a molecular level and within crystalline systems. An understanding of symmetry is essential in discussions of molecular spectroscopy and calculations of molecular properties. A discussion of *crystal symmetry* is not included in this book, and we introduce only *molecular symmetry*. For qualitative purposes, it is sufficient to refer to the shape of a molecule using terms such as tetrahedral, octahedral or square planar. However, the common use of these descriptors is not always precise, e.g. consider the structures of BF_3 , **3.1**, and BF_2H , **3.2**, both of which are planar. A molecule of BF_3 is correctly described as being trigonal planar, since its symmetry properties are fully consistent with this description; all the F–B–F bond angles are 120° and the B–F bond distances are all identical (131 pm). It is correct to say that the boron centre in BF_2H , **3.2**, is in a *pseudo-trigonal planar* environment but the molecular symmetry properties are not the same as those of BF_3 . The F–B–F bond angle in BF_2H is smaller than the two H–B–F angles, and the B–H bond is shorter (119 pm) than the B–F bonds (131 pm).



(3.1)



(3.2)

The descriptor *symmetrical* implies that a species possesses a number of indistinguishable configurations. When structure **3.1** is rotated in the plane of the paper

through 120° , the resulting structure is indistinguishable from the first; another 120° rotation results in a third indistinguishable molecular orientation (Fig. 3.1). This is *not* true if we carry out the same rotational operations on BF_2H .

Group theory is the mathematical treatment of symmetry. In this chapter, we introduce the fundamental language of group theory (*symmetry operator*, *symmetry element*, *point group* and *character table*). The chapter does not set out to give a comprehensive survey of molecular symmetry, but rather to introduce some common terminology and its meaning. We include an introduction to the vibrational spectra of simple inorganic molecules and show how to use this technique to distinguish between possible structures for XY_2 , XY_3 and XY_4 molecules. Complete normal coordinate analysis of such species is beyond the remit of this book.

3.2 Symmetry operations and symmetry elements

In Fig. 3.1, we applied 120° rotations to BF_3 and saw that each rotation generated a representation of the molecule that was indistinguishable from the first. Each rotation is an example of a *symmetry operation*.

A **symmetry operation** is an operation performed on an object which leaves it in a configuration that is indistinguishable from, and superimposable on, the original configuration.

The rotations described in Fig. 3.1 are performed about an axis perpendicular to the plane of the paper and passing through the boron atom; the axis is an example of a *symmetry element*.

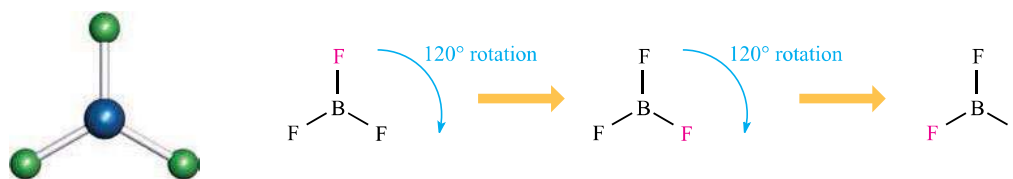


Fig. 3.1 Rotation of the trigonal planar BF_3 molecule through 120° generates a representation of the structure that is indistinguishable from the first; one F atom is marked in red simply as a label. A second 120° rotation gives another indistinguishable structural representation.

A symmetry operation is carried out with respect to points, lines or planes, the latter being the *symmetry elements*.

Rotation about an n -fold axis of symmetry

The symmetry operation of rotation about an n -fold axis (the symmetry element) is denoted by the symbol C_n , in which the angle of rotation is $\frac{360^\circ}{n}$; n is an integer, e.g. 2, 3 or 4. Applying this notation to the BF_3 molecule in Fig. 3.1 gives a value of $n = 3$ (eq. 3.1), and therefore we say that the BF_3 molecule contains a C_3 rotation axis. In this case, the axis lies perpendicular to the plane containing the molecule.

$$\text{Angle of rotation} = 120^\circ = \frac{360^\circ}{n} \quad (3.1)$$

In addition, BF_3 also contains three 2-fold (C_2) rotation axes, each coincident with a B–F bond as shown in Fig. 3.2.

If a molecule possesses more than one type of n -axis, the axis of highest value of n is called the *principal axis*; it is the axis of *highest molecular symmetry*. For example, in BF_3 , the C_3 axis is the principal axis.

In some molecules, rotation axes of lower orders than the principal axis may be coincident with the principal axis. For example, in square planar XeF_4 , the principal axis is a C_4 axis but this also coincides with a C_2 axis (see Fig. 3.4).

Where a molecule contains more than one type of C_n axis with the same value of n , they are distinguished by using prime marks, e.g. C_2 , C_2' and C_2'' . We return to this in the discussion of XeF_4 (see Fig. 3.4).

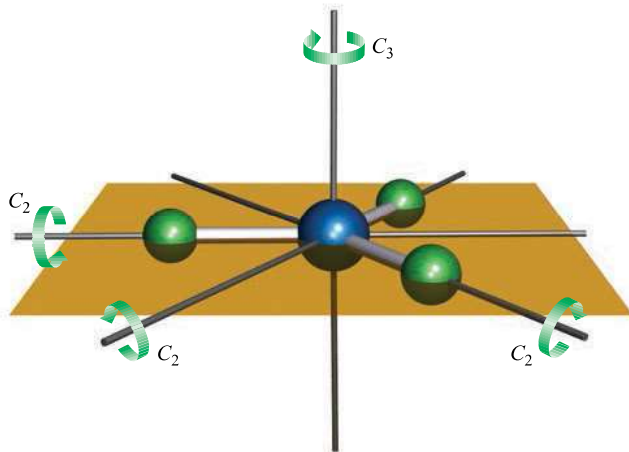


Fig. 3.2 The 3-fold (C_3) and three 2-fold (C_2) axes of symmetry possessed by the trigonal planar BF_3 molecule.

Self-study exercises

- Each of the following contains a 6-membered ring: benzene, borazine (see Fig. 13.23), pyridine and S_6 (see Box 1.1). Explain why only benzene contains a 6-fold principal rotation axis.
- Among the following, why does only XeF_4 contain a 4-fold principal rotation axis: CF_4 , SF_4 , $[\text{BF}_4]^-$ and XeF_4 ?
- Draw the structure of $[\text{XeF}_5]^-$. On the diagram, mark the C_5 axis. The ion contains five C_2 axes. Where are these axes? [Ans. for structure, see worked example 2.7]
- Look at the structure of B_5H_9 in Fig. 13.29a. Where is the C_4 axis in this molecule?

Reflection through a plane of symmetry (mirror plane)

If reflection of all parts of a molecule through a plane produces an indistinguishable configuration, the plane is a *plane of symmetry*; the symmetry operation is one of reflection and the symmetry element is the mirror plane (denoted by σ). For BF_3 , the plane containing the molecular framework (the brown plane shown in Fig. 3.2) is a *mirror plane*. In this case, the plane lies perpendicular to the vertical principal axis and is denoted by the symbol σ_h .

The framework of atoms in a linear, bent or planar molecule can always be drawn in a plane, but this plane can be labelled σ_h *only* if the molecule possesses a C_n axis *perpendicular* to the plane. If the plane *contains* the principal axis, it is labelled σ_v . Consider the H_2O molecule. This possesses a C_2 axis (Fig. 3.3) but it also contains *two* mirror planes, one containing the H_2O framework, and one perpendicular to it. Each plane contains the principal axis of rotation and so may be denoted as σ_v but in order to distinguish between them, we use the notations σ_v and σ_v' . The σ_v label refers to the plane that bisects the H–O–H bond angle and the σ_v' label refers to the plane in which the molecule lies.

A special type of σ plane which contains the principal rotation axis, but which bisects the angle between two adjacent 2-fold axes, is labelled σ_d . A square planar

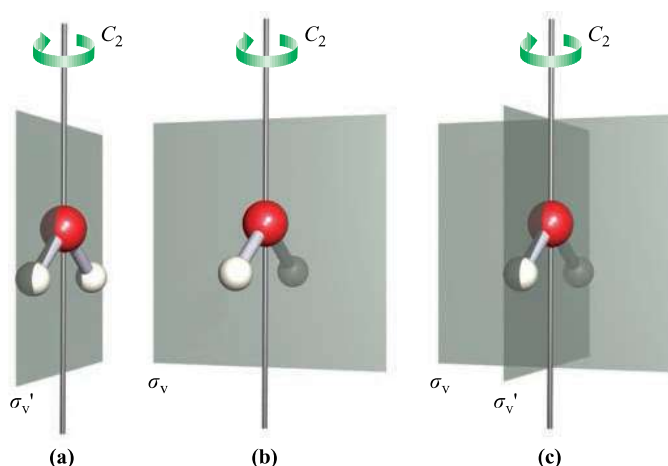


Fig. 3.3 The H_2O molecule possesses one C_2 axis and two mirror planes. (a) The C_2 axis and the plane of symmetry that contains the H_2O molecule. (b) The C_2 axis and the plane of symmetry that is perpendicular to the plane of the H_2O molecule. (c) Planes of symmetry in a molecule are often shown together on one diagram; this representation for H_2O combines diagrams (a) and (b).

molecule such as XeF_4 provides an example. Figure 3.4a shows that XeF_4 contains a C_4 axis (the principal axis) and perpendicular to this is the σ_h plane in which the molecule lies. Coincident with the C_4 axis is a C_2 axis. Within the plane of the molecule, there are two sets of C_2 axes. One type (the C_2' axis) coincides with F-Xe-F bonds, while the second type (the C_2'' axis) bisects the F-Xe-F 90° angle (Fig. 3.4). We can now define two sets of mirror planes: one type (σ_v) contains the principal axis and a C_2' axis (Fig. 3.4b), while the second type (σ_d) contains the principal axis and a C_2'' axis (Fig. 3.4c). Each σ_d plane bisects the angle between two C_2' axes.

In the notation for planes of symmetry, σ , the subscripts h, v and d stand for horizontal, vertical and dihedral respectively.

Self-study exercises

1. N_2O_4 is planar (Fig. 15.15). Show that it possesses three planes of symmetry.

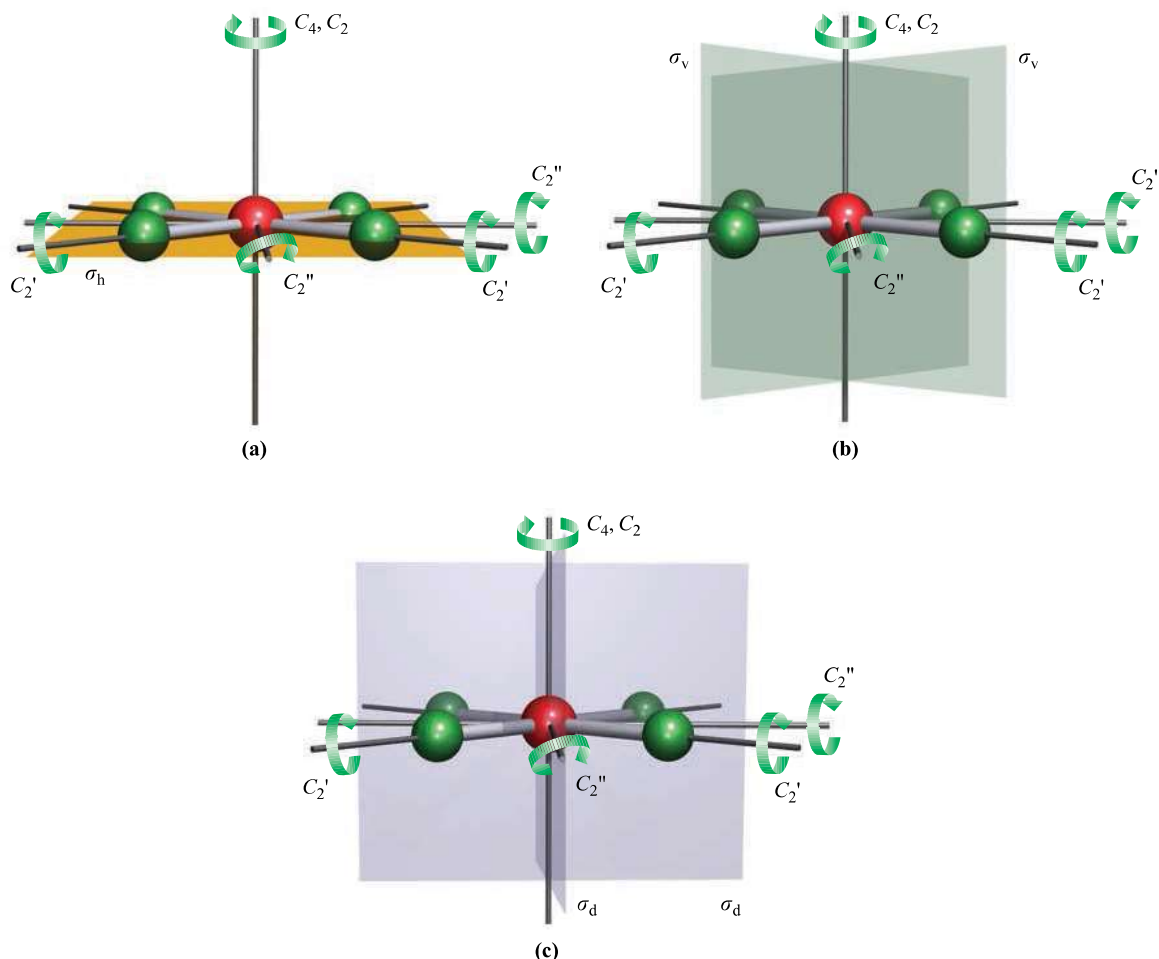
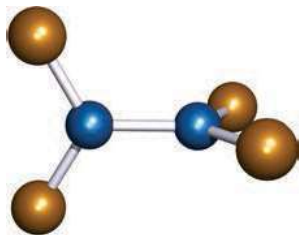


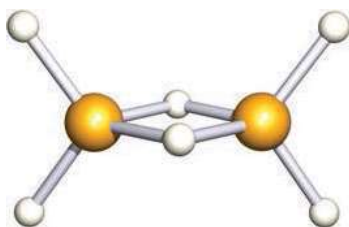
Fig. 3.4 The square planar molecule XeF_4 . (a) One C_2 axis coincides with the principal (C_4) axis; the molecule lies in a σ_h plane which contains two C_2' and two C_2'' axes. (b) Each of the two σ_v planes contains the C_4 axis and one C_2' axis. (c) Each of the two σ_d planes contains the C_4 axis and one C_2'' axis.

2. B_2Br_4 has the following staggered structure:



Show that B_2Br_4 has one less plane of symmetry than B_2F_4 which is planar.

3. Ga_2H_6 has the following structure in the gas phase:

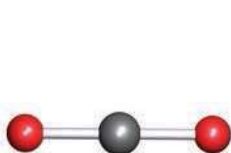


Show that it possesses three planes of symmetry.

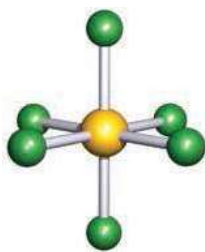
4. Show that the planes of symmetry in benzene are one σ_h , three σ_v and three σ_d .

Reflection through a centre of symmetry (inversion centre)

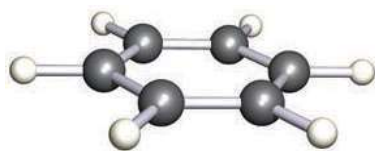
If reflection of *all* parts of a molecule through the centre of the molecule produces an indistinguishable configuration, the centre is a *centre of symmetry*. It is also called a *centre of inversion* (see also Box 2.1) and is designated by the symbol i . Each of the molecules CO_2 (3.3), *trans*- N_2F_2 (see worked example 3.1), SF_6 (3.4) and benzene (3.5) possesses a centre of symmetry, but H_2S (3.6), *cis*- N_2F_2 (3.7) and SiH_4 (3.8) do not.



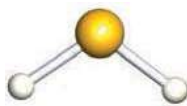
(3.3)



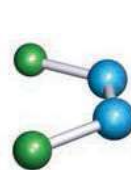
(3.4)



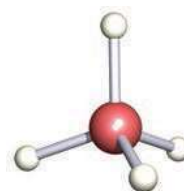
(3.5)



(3.6)



(3.7)



(3.8)

Self-study exercises

1. Draw the structures of each of the following species and confirm that each possesses a centre of symmetry: CS_2 , $[\text{PF}_6]^-$, XeF_4 , I_2 , $[\text{ICl}_2]^-$.
2. $[\text{PtCl}_4]^{2-}$ has a centre of symmetry, but $[\text{CoCl}_4]^{2-}$ does not. One is square planar and the other is tetrahedral. Which is which?
3. Why does CO_2 possess an inversion centre, but NO_2 does not?
4. CS_2 and HCN are both linear. Explain why CS_2 possesses a centre of symmetry whereas HCN does not.

Rotation about an axis, followed by reflection through a plane perpendicular to this axis

If rotation through $\frac{360^\circ}{n}$ about an axis, followed by reflection through a plane perpendicular to that axis, yields an indistinguishable configuration, the axis is an *n-fold rotation-reflection axis*, also called an *n-fold improper rotation axis*. It is denoted by the symbol S_n . Tetrahedral species of the type XY_4 (all Y groups must be equivalent) possess three S_4 axes, and the operation of one S_4 rotation-reflection in the CH_4 molecule is illustrated in Fig. 3.5.

Self-study exercises

1. Explain why BF_3 possesses an S_3 axis, but NF_3 does not.
2. C_2H_6 in a staggered conformation possesses an S_6 axis. Show that this axis lies along the C–C bond.
3. Figure 3.5 shows one of the S_4 axes in CH_4 . On going from CH_4 to CH_2Cl_2 , are the S_4 axes retained?

Identity operator

All objects can be operated upon by the identity operator E . This is the simplest operator (although it may not be easy to appreciate why we identify such an operator!) and effectively identifies the molecular configuration. The operator E leaves the molecule unchanged.

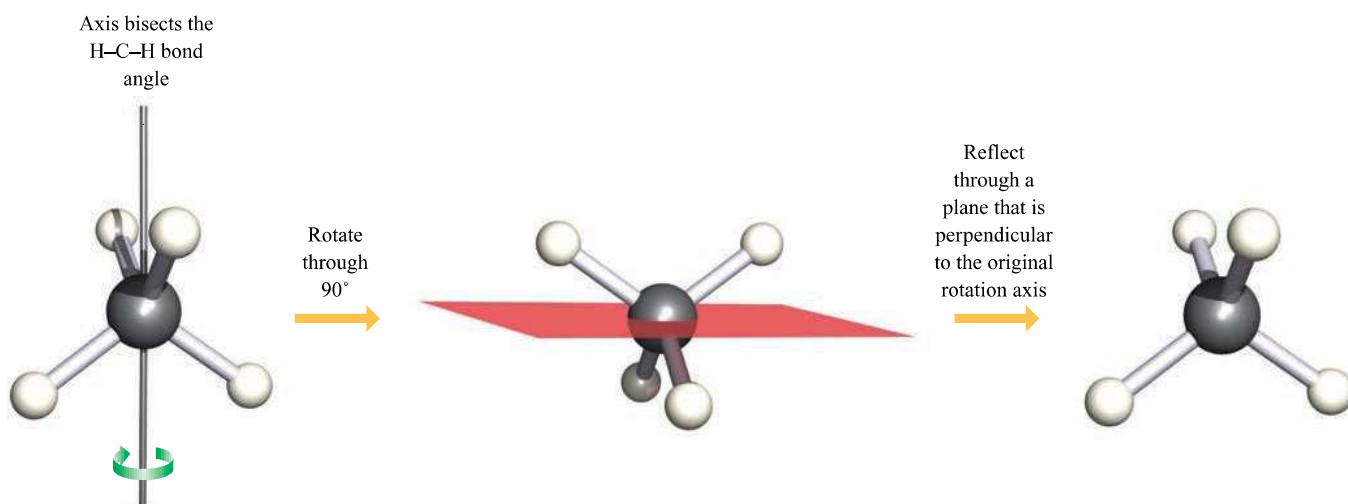
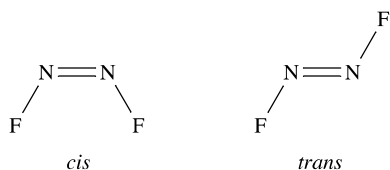


Fig. 3.5 An improper rotation (or rotation-reflection), S_n , involves rotation about $\frac{360^\circ}{n}$ followed by reflection through a plane that is perpendicular to the rotation axis. The diagram illustrates the operation about one of the S_4 axes in CH_4 ; three S_4 operations are possible for the CH_4 molecule. [Exercise: where are the three rotation axes for the three S_4 operations in CH_4 ?]

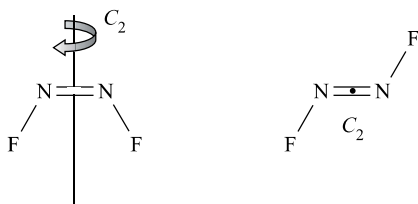
Worked example 3.1 Symmetry properties of *cis*- and *trans*- N_2F_2

How do the rotation axes and planes of symmetry in *cis*- and *trans*- N_2F_2 differ?

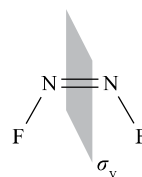
First draw the structures of *cis*- and *trans*- N_2F_2 ; both are planar molecules.



1. The identity operator E applies to each isomer.
2. Each isomer possesses a plane of symmetry which contains the molecular framework. However, their labels differ (see point 5 below).
3. The *cis*-isomer contains a C_2 axis which lies in the plane of the molecule, but the *trans*-isomer contains a C_2 axis which bisects the N-N bond and is perpendicular to the plane of the molecule.



4. The *cis*- (but not the *trans*-) isomer contains a mirror plane, σ_v , lying perpendicular to the plane of the molecule and bisecting the N-N bond:



5. The consequence of the different types of C_2 axes, and the presence of the σ_v plane in the *cis*-isomer, is that the symmetry planes containing the *cis*- and *trans*- N_2F_2 molecular frameworks are labelled σ_v' and σ_h respectively.

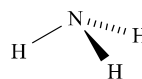
Self-study exercises

1. How do the rotation axes and planes of symmetry in *Z*- and *E*- $\text{CFH}=\text{CFH}$ differ?
2. How many planes of symmetry do (a) $\text{F}_2\text{C}=\text{O}$, (b) $\text{ClFC}=\text{O}$ and (c) $[\text{HCO}_2]^-$ possess?
[Ans. (a) 2; (b) 1; (c) 2]

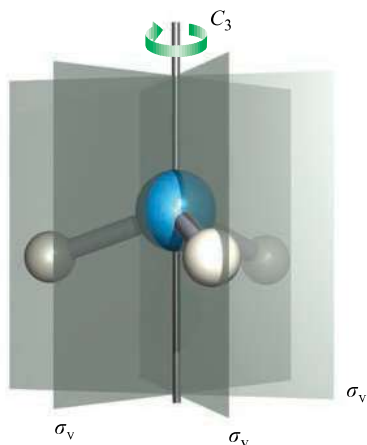
Worked example 3.2 Symmetry elements in NH_3

The symmetry elements for NH_3 are E , C_3 and $3\sigma_v$. (a) Draw the structure of NH_3 . (b) What is the meaning of the E operator? (c) Draw a diagram to show the symmetry elements.

- (a) The molecule is trigonal pyramidal.



- (b) The E operator is the identity operator and it leaves the molecule unchanged.
- (c) The C_3 axis passes through the N atom, perpendicular to a plane containing the three H atoms. Each σ_v plane contains one N–H bond and bisects the opposite H–N–H bond angle.



Self-study exercises

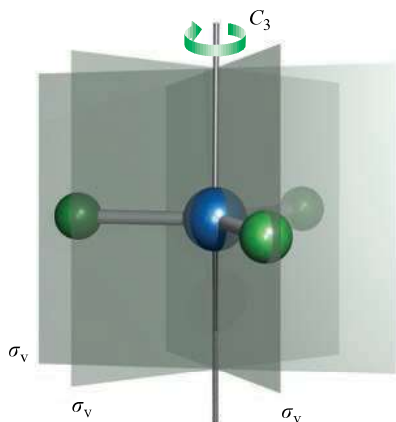
- What symmetry elements are lost in going from NH_3 to NH_2Cl ? [Ans. C_3 ; two σ_v]
- Compare the symmetry elements possessed by NH_3 , NH_2Cl , NHCl_2 and NCl_3 .
- Draw a diagram to show the symmetry elements of NClF_2 . [Ans. Show one σ plane; only other operator is E]

Worked example 3.3 Trigonal planar BCl_3 versus trigonal pyramidal PCl_3

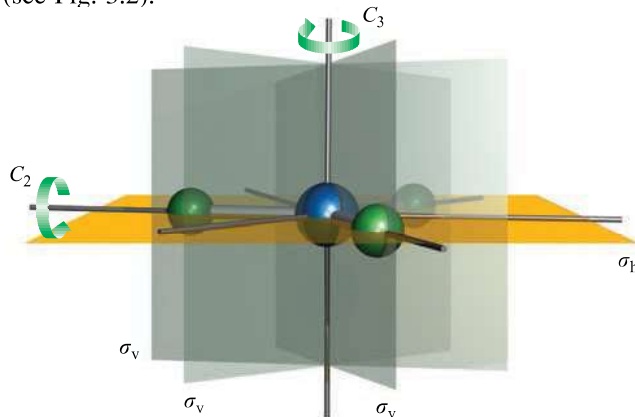
What symmetry elements do BCl_3 and PCl_3 (a) have in common and (b) not have in common?

PCl_3 is trigonal pyramidal (use the VSEPR model) and so possesses the same symmetry elements as NH_3 in worked example 3.2. These are E , C_3 and $3\sigma_v$.

BCl_3 is trigonal planar (use VSEPR) and possesses all the above symmetry elements:



In addition, BCl_3 contains a σ_h plane and three C_2 axes (see Fig. 3.2).



Rotation through 120° about the C_3 axis, followed by reflection through the plane perpendicular to this axis (the σ_h plane), generates a molecular configuration indistinguishable from the first – this is an improper rotation S_3 .

Conclusion

The symmetry elements that BCl_3 and PCl_3 have in common are E , C_3 and $3\sigma_v$. The symmetry elements possessed by BCl_3 but not by PCl_3 are σ_h , $3C_2$ and S_3 .

Self-study exercises

- Show that BF_3 and $\text{F}_2\text{C}=\text{O}$ have the following symmetry elements in common: E , two mirror planes, one C_2 .
- How do the symmetry elements of ClF_3 and BF_3 differ? [Ans: BF_3 , as for BCl_3 above; ClF_3 , E , σ_v' , σ_v , C_2]

3.3 Successive operations

As we have seen in Section 3.2, a particular symbol is used to denote a specific symmetry element. To say that NH_3 possesses a C_3 axis tells us that we can rotate the molecule through 120° and end up with a molecular configuration that is indistinguishable from the first. However, it takes three such operations to give a configuration of the NH_3 molecule that *exactly* coincides with the first. The three separate 120° rotations are identified by using the notation in Fig. 3.6. We cannot *actually* distinguish between the three H atoms, but for clarity they are labelled H(1), H(2) and H(3) in the figure. Since the third rotation, C_3^3 , returns the NH_3 molecule to its initial configuration, we can write eq. 3.2, or, in general, eq. 3.3.

$$C_3^3 = E \quad (3.2)$$

$$C_n^n = E \quad (3.3)$$

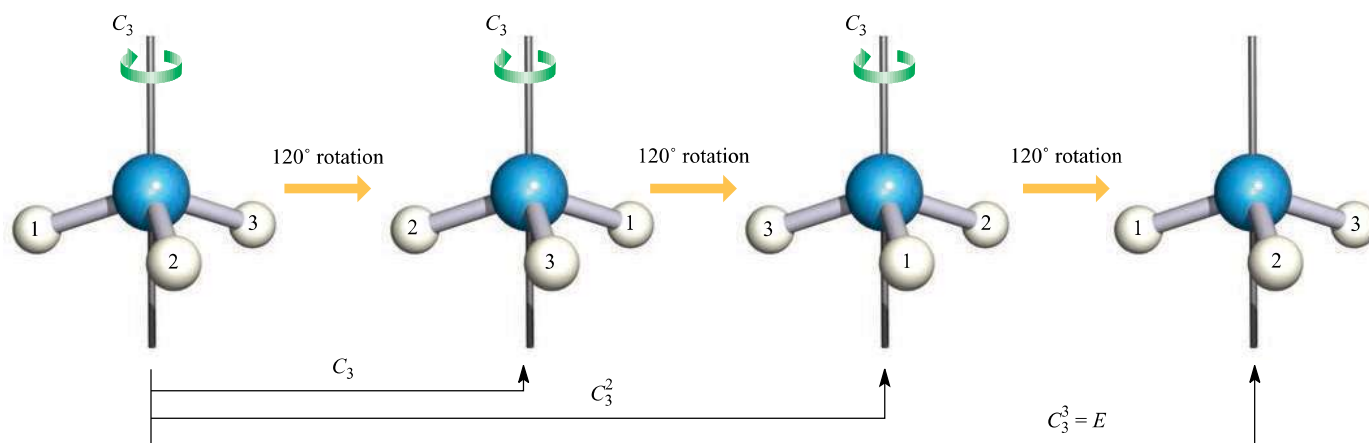


Fig. 3.6 Successive C_3 rotations in NH_3 are distinguished using the notation C_3 (or C_3^1), C_3^2 and C_3^3 . The effect of the last operation is the same as that of the identity operator acting on NH_3 in the initial configuration.

Similar statements can be written to show the combined effects of successive operations. For example, in planar BCl_3 , the S_3 improper axis of rotation corresponds to rotation about the C_3 axis followed by reflection through the σ_h plane. This can be written in the form of eq. 3.4.

$$S_3 = C_3 \times \sigma_h \quad (3.4)$$

Self-study exercises

1. $[\text{PtCl}_4]^{2-}$ is square planar; to what rotational operation is C_4^2 equivalent?
2. Draw a diagram to illustrate what the notation C_6^4 means with respect to rotational operations in benzene.

3.4 Point groups

The number and nature of the symmetry elements of a given molecule are conveniently denoted by its *point group*, and give rise to labels such as C_2 , C_{3v} , D_{3h} , D_{2d} , T_d , O_h or I_h . These point groups belong to the classes of C groups, D groups and special groups, the latter containing groups that possess special symmetries, i.e. tetrahedral, octahedral and icosahedral.

To describe the symmetry of a molecule in terms of one symmetry element (e.g. a rotation axis) provides information only about this property. Each of BF_3 and NH_3 possesses a 3-fold axis of symmetry, but their structures and overall symmetries are different: BF_3 is trigonal planar and NH_3 is trigonal pyramidal. On the other hand, if we describe the symmetries of these molecules in terms of their respective point groups (D_{3h} and C_{3v}), we are providing information about *all* their symmetry elements.

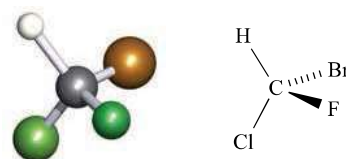
Before we look at some representative point groups, we emphasize that it is not essential to memorize the symmetry elements of a particular point group. These are listed

in *character tables* (see Sections 3.5, 5.4 and 5.5, and Appendix 3) which are widely available.

Table 3.1 summarizes the most important classes of point group and gives their characteristic types of symmetry elements; E is, of course, common to every group. Some particular features of significance are given below.

C_1 point group

Molecules that appear to have no symmetry at all, e.g. **3.9**, must possess the symmetry element E and effectively possess at least one C_1 axis of rotation. They therefore belong to the C_1 point group, although since $C_1 = E$, the rotational symmetry operation is ignored when we list the symmetry elements of this point group.



(3.9)

$C_{\infty v}$ point group

C_{∞} signifies the presence of an ∞ -fold axis of rotation, i.e. that possessed by a linear molecule (Fig. 3.7); for the molecular species to belong to the $C_{\infty v}$ point group, it must also possess an infinite number of σ_v planes but *no* σ_h plane or inversion centre. These criteria are met by asymmetrical diatomics such as HF , CO and $[\text{CN}]^-$ (Fig. 3.7a), and linear polyatomics (throughout this book, polyatomic is used to mean a species containing three or more atoms) that do not possess a centre of symmetry, e.g. OCS and HCN .

$D_{\infty h}$ point group

Symmetrical diatomics (e.g. H_2 , $[\text{O}_2]^{2-}$) and linear polyatomics that contain a centre of symmetry (e.g. $[\text{N}_3]^-$, CO_2 , $\text{HC}\equiv\text{CH}$) possess a σ_h plane in addition to a C_{∞}

Table 3.1 Characteristic symmetry elements of some important classes of point groups. The characteristic symmetry elements of the T_d , O_h and I_h are omitted because the point groups are readily identified (see Figs. 3.8 and 3.9). No distinction is made in this table between σ_v and σ_d planes of symmetry. For complete lists of symmetry elements, character tables (Appendix 3) should be consulted.

Point group	Characteristic symmetry elements	Comments
C_s	E , one σ plane	
C_i	E , inversion centre	
C_n	E , one (principal) n -fold axis	
C_{nv}	E , one (principal) n -fold axis, n σ_v planes	
C_{nh}	E , one (principal) n -fold axis, one σ_h plane, one S_n -fold axis which is coincident with the C_n axis	The S_n axis necessarily follows from the C_n axis and σ_h plane. For $n = 2, 4$ or 6 , there is also an inversion centre.
D_{nh}	E , one (principal) n -fold axis, n C_2 axes, one σ_h plane, n σ_v planes, one S_n -fold axis	The S_n axis necessarily follows from the C_n axis and σ_h plane. For $n = 2, 4$ or 6 , there is also an inversion centre.
D_{nd}	E , one (principal) n -fold axis, n C_2 axes, n σ_v planes, one S_{2n} -fold axis	For $n = 3$ or 5 , there is also an inversion centre.
T_d		Tetrahedral
O_h		Octahedral
I_h		Icosahedral

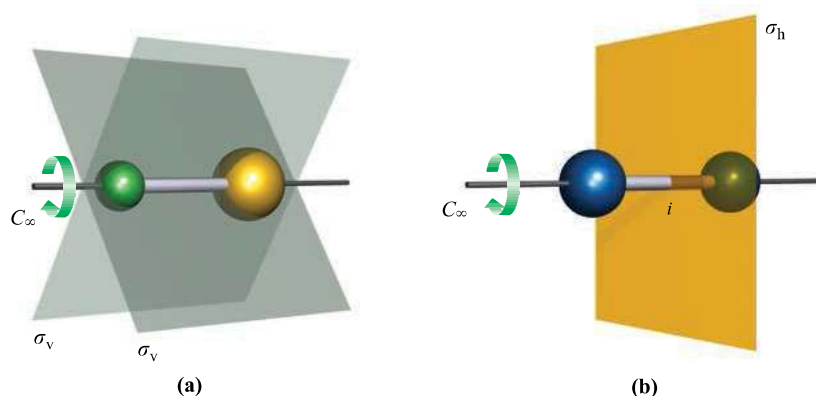


Fig. 3.7 Linear molecular species can be classified according to whether they possess a centre of symmetry (inversion centre) or not. All linear species possess a C_∞ axis of rotation and an infinite number of σ_v planes; in (a), two such planes are shown and these planes are omitted from (b) for clarity. Diagram (a) shows an asymmetrical diatomic belonging to the point group $C_{\infty v}$, and (b) shows a symmetrical diatomic belonging to the point group $D_{\infty h}$.

axis and an infinite number of σ_v planes (Fig. 3.7). These species belong to the $D_{\infty h}$ point group.

T_d , O_h or I_h point groups

Molecular species that belong to the T_d , O_h or I_h point groups (Fig. 3.8) possess many symmetry elements, although it is seldom necessary to identify them all before the appropriate point group can be assigned. Species with tetrahedral symmetry include SiF_4 , $[\text{ClO}_4]^-$, $[\text{CoCl}_4]^{2-}$, $[\text{NH}_4]^+$, P_4 (Fig. 3.9a) and B_4Cl_4 (Fig. 3.9b). Those with octahedral symmetry include SF_6 , $[\text{PF}_6]^-$, $\text{W}(\text{CO})_6$

(Fig. 3.9c) and $[\text{Fe}(\text{CN})_6]^{3-}$. There is no centre of symmetry in a tetrahedron but there is one in an octahedron, and this distinction has consequences with regard to the observed electronic spectra of tetrahedral and octahedral metal complexes (see Section 20.7). Members of the icosahedral point group are uncommon, e.g. $[\text{B}_{12}\text{H}_{12}]^{2-}$ (Fig. 3.9d).

Determining the point group of a molecule or molecular ion

The application of a *systematic* approach to the assignment of a point group is essential, otherwise there is the risk that

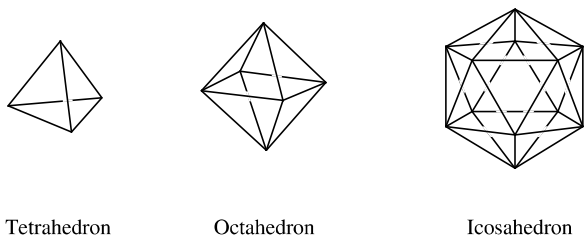


Fig. 3.8 The tetrahedron (T_d symmetry), octahedron (O_h symmetry) and icosahedron (I_h symmetry) possess 4, 6 and 12 vertices respectively, and 4, 8 and 20 equilateral-triangular faces respectively.

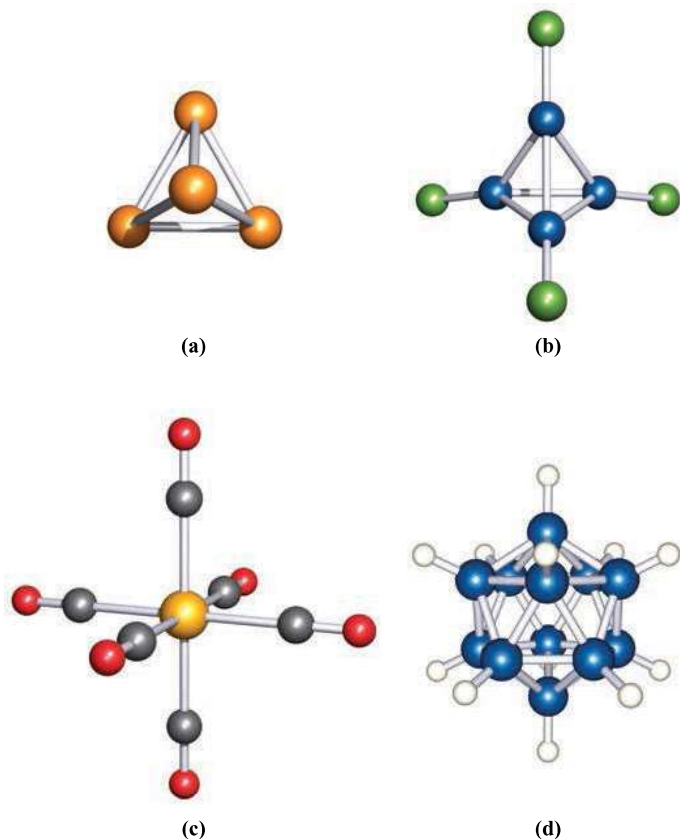


Fig. 3.9 The molecular structures of (a) P_4 , (b) B_4Cl_4 (the B atoms are shown in blue), (c) $[W(CO)_6]$ (the W atom is shown in yellow and the C atoms in grey) and (d) $[B_{12}H_{12}]^{2-}$ (the B atoms are shown in blue).

symmetry elements will be missed with the consequence that an incorrect assignment is made. Figure 3.10 shows a procedure that may be adopted; some of the less common point groups (e.g. S_n , T , O) are omitted from the scheme. Notice that it is *not* necessary to find all the symmetry elements (e.g. improper axes) in order to determine the point group.

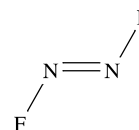
We illustrate the application of Fig. 3.10 in four worked examples, with an additional example in Section 3.8. Before assigning a point group to a molecule, its structure must be determined by, for example, microwave

spectroscopy, or X-ray, electron or neutron diffraction methods (see Section 4.10).

Worked example 3.4 Point group assignments: 1

Determine the point group of $trans\text{-N}_2\text{F}_2$.

First draw the structure.



Apply the strategy shown in Fig. 3.10:

START $\Rightarrow$

- | | |
|---|--|
| Is the molecule linear? | No |
| Does $trans\text{-N}_2\text{F}_2$ have T_d , O_h or I_h symmetry? | No |
| Is there a C_n axis? | Yes; a C_2 axis perpendicular to the plane of the paper and passing through the midpoint of the N–N bond |
| Are there two C_2 axes perpendicular to the principal axis? | No |
| Is there a σ_h plane (perpendicular to the principal axis)? | Yes |

$\Rightarrow$ **STOP**

The point group is C_{2h} .

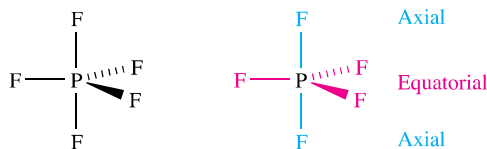
Self-study exercises

1. Show that the point group of $cis\text{-N}_2\text{F}_2$ is C_{2v} .
2. Show that the point group of $E\text{-CHCl=CHCl}$ is C_{2h} .

Worked example 3.5 Point group assignments: 2

Determine the point group of PF_5 .

First, draw the structure.



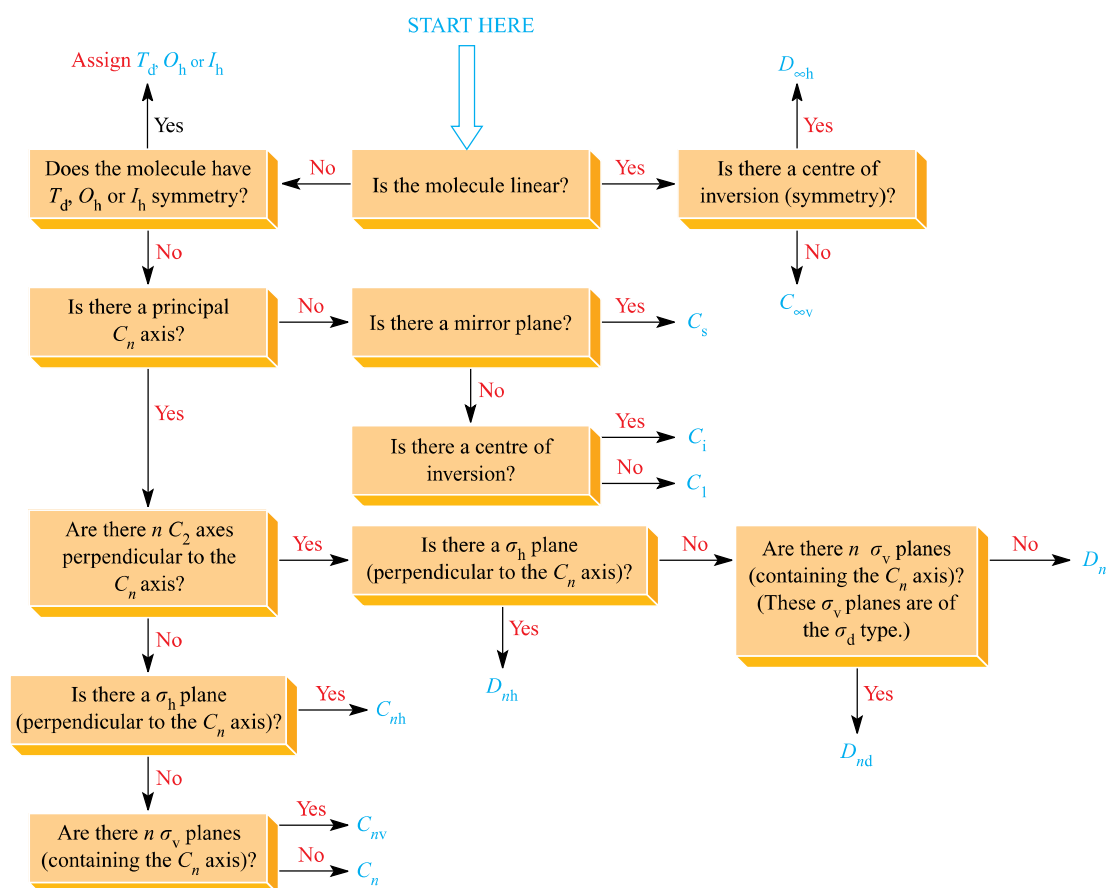


Fig. 3.10 Scheme for assigning point groups of molecules and molecular ions. Apart from the cases of $n = 1$ or ∞ , n most commonly has values of 2, 3, 4, 5 or 6.

In the trigonal bipyramidal arrangement, the three equatorial F atoms are equivalent, and the two axial F atoms are equivalent.

Apply the strategy shown in Fig. 3.10:

START $\Rightarrow$

Is the molecule linear?	No
Does PF_5 have T_d , O_h or I_h symmetry?	No
Is there a C_n axis?	Yes; a C_3 axis containing the P and two axial F atoms

Are there three C_2 axes perpendicular to the principal axis?	Yes; each lies along an equatorial P–F bond
Is there a σ_h plane (perpendicular to the principal axis)?	Yes; it contains the P and three equatorial F atoms.

$\Rightarrow$ **STOP**

The point group is D_{3h} .

Self-study exercises

1. Show that BF_3 belongs to the D_{3h} point group.
2. Show that OF_2 belongs to the C_{2v} point group.
3. Show that BF_2Br belongs to the C_{2v} point group.

Worked example 3.6 Point group assignments: 3

To what point group does POCl_3 belong?

The structure of POCl_3 is:

